

EXPERIMENT NO: 3

ROLL NO:240701288

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Survey Report on Emerging Trends in Low-Code Application Development

1. Aim:

To study the emerging trends in **low-code application development**, understand how low-code platforms differ from traditional application development, analyze their benefits and challenges, and compare commonly used low-code platforms.

2. Traditional Application Development vs Emerging Trends in Low-Code Application Development:

2.1 Traditional Application Development:-

Traditional application development involves building applications mainly through manual coding. Developers write code for the interface, business logic, database, testing, and deployment. It provides high flexibility and customization but requires skilled programmers and generally takes more time and effort to develop and maintain.

2.2 Low-Code Application Development:-

Low-code development uses **visual tools, drag-and-drop components, pre-built modules, and reusable elements** to create applications with minimal manual coding. It supports faster development and allows both professional developers and business users to participate in application development. Custom coding can still be used when advanced features are required.

2.3 Emerging Trends in Low-Code Application Development:-

The important emerging trends are:

- **AI-Assisted Development:** AI helps generate code, formulas, workflows, and application components.
- **Workflow Automation:** Business processes such as approvals, notifications, and record updates can be automated.
- **Citizen Development:** Non-professional users can build applications using visual development tools.
- **Cloud-Based Development:** Applications can be developed, deployed, and managed through cloud platforms.
- **API Integration:** Applications can easily connect with databases, APIs, and external services.
- **Reusable Components:** Templates and components can be reused to reduce development effort.
- **Rapid Prototyping:** Working prototypes can be created quickly and improved based on feedback.
- **Mobile Development:** Low-code platforms support responsive and mobile applications.
- **Data Analytics:** Reports, dashboards, and visualizations can be integrated into applications.

2.4 Summary

Traditional development depends mainly on manual coding, while low-code development emphasizes **visual development, automation, and reusable components**. With emerging technologies such as AI, cloud computing, API integration, and analytics, low-code platforms are making application development faster and more accessible.

3. Benefits and Challenges in Low-Code Development:

3.1 Benefits

1. Faster Development

Drag-and-drop components and pre-built functionality reduce the amount of manual coding required.

2. Reduced Development Effort

Developers can use existing components, integrations, and workflows rather than implementing everything from scratch.

3. Easy Prototyping

A working prototype can be created quickly, making it easier to demonstrate and refine application requirements.

4. Increased Productivity

Developers can concentrate on complex business logic while the platform handles common application functions.

5. Citizen Development

Employees with limited programming knowledge can develop applications for simple business requirements.

6. Workflow Automation

Processes such as notifications, approvals, data validation, and record updates can be automated.

7. Easier Maintenance

Visual components and centralized platform features can simplify application maintenance.

8. Integration Support

Many low-code platforms provide ready-made connectors for databases, APIs, cloud services, and business applications.

3.2 Challenges

1. Limited Customization

Highly specialized requirements may be difficult to implement if they are not supported by the platform.

2. Vendor Lock-in

Applications can become dependent on a particular platform's features, pricing, and infrastructure.

3. Security Concerns

Applications must be configured carefully to protect sensitive information and control user access.

4. Performance Limitations

Very complex or high-scale applications may require more control than a low-code platform provides.

5. Learning the Platform

Although coding requirements are reduced, users still need to understand the platform's components, workflows, and rules.

6. Cost

Advanced features, integrations, storage, and larger numbers of users may require paid plans.

7. Dependency on Platform Updates

Changes made by the platform provider can affect application functionality.

8. Complex Applications May Still Need Coding

Low-code does not completely eliminate programming. Custom functions or advanced requirements may still require conventional code.

4. Comparison of Existing Low-Code Platforms:

Comparison of Existing Low-Code Platforms

Feature	 Zoho Creator	 Microsoft Power Apps	 OutSystems	 Mendix	 Salesforce Platform
Developer	Zoho	Microsoft	OutSystems	Mendix	Salesforce
Development Approach	Visual + low-code	Low-code + visual development	Low-code	Low-code	Low-code
Form Building	Drag-and-drop	Visual controls	Visual UI development	Visual development	Visual development
Workflow Automation	Supported	Supported through Power Automate	Supported	Supported	Supported
Database	Built-in application database	Dataverse and connectors	Built-in/data integration options	Supported	Salesforce data platform
API Integration	Supported	Extensive connectors	Supported	Supported	Supported
Mobile Development	Supported	Supported	Supported	Supported	Supported
AI Features	AI-assisted capabilities	AI/Copilot capabilities	AI-assisted development capabilities	AI-assisted development capabilities	AI-assisted capabilities
Custom Coding	Deluge and other supported options	Power Fx and extensions	Supported	Supported	Apex and other development tools
Main Strength	Business applications and rapid development	Microsoft ecosystem integration	Enterprise application development	Enterprise low-code development	CRM and business applications
Typical Users	Businesses, developers, citizen developers	Businesses and developers	Enterprise teams	Enterprise teams	CRM/business users and developers

5.Application used in the Experiment :

Platform: Zoho Creator

Application: Patient Appointment & Scheduling Management

Form: Appointment_Booking_Form

Report: Appointment_Booking_Report

The application uses the **13 Basic Fields** in Zoho Creator to collect patient personal details, contact information, medical details, preferred appointment schedule, and treatment consent. The submitted information is stored in the appointment report, allowing the front desk to review and manage appointment requests.

FORM:

The screenshot displays a web browser window with the URL `creatorapp.zoho.in/lipika.j.2024.cse_rajalak421/patient-appointment-booking/#Form:Appointment_Booking_Form`. The browser's address bar shows the Zoho Creator logo, navigation icons, and a user profile labeled 'School'. The page title is 'Patient Appointment Booking'. A sidebar on the left contains a menu with 'Patient Appointment...' (highlighted), 'Appointment_Booking_Form', and 'Appointment_Booking_Form Report'. The main content area is titled 'Appointment_Booking_Form' and contains the following fields:

- Patient_Name:** Two text input fields for 'First Name' and 'Last Name'.
- Age:** A text input field with a placeholder '#####'.
- Gender:** Three radio button options: 'Male', 'Female', and 'Others'.
- Email Address:** A text input field.
- Contact Number:** A text input field with a country code dropdown set to '+91' and a pre-filled number '81234 56789'.
- Residential Address:** A text input field with a label 'Address Line 1' below it.
- Referred By:** A text input field.
- Department:** A dropdown menu with '-Select-' as the current selection.
- Existing Conditions:** A dropdown menu with '-Select-' as the current selection.
- Symptoms Description:** A large text area for input.
- Preferred Appointment Date:** A text input field with a placeholder 'dd-MMM-yyyy'.
- Preferred Appointment Time:** A text input field with a placeholder 'HH:mm:ss'.
- Consent_For_Treatment:** A checkbox labeled 'I consent to examination and treatment'.

A blue 'Submit' button is located at the bottom right of the form.

Patient Appointment Booking

Search this tab with Google Lens

Appointment_Booking_Form

Appointment_Booking_Form

Appointment_Booking_Form Report

Patient Appointment...

Appointment_Booking_Form

Patient_Name

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First NameLast Name

Age

20

Gender

☐ Male

☒ Female

☐ Others

Email Address

lipika.j.2024.cse@rajalakshmi.edu.in

Contact Number

+91

9080225657

Residential Address

19/2B Grace garden Royapuram

Address Line 1

Referred By

DR.SHREYESH

Department

Dermatology

×

▼

Existing Conditions

Hypertension

×

Symptoms Description

SKIN REDNESS AND DRYNESS

Preferred Appointment Date

03-Jun-2026

Preferred Appointment Time

08:47:36

Consent_For_Treatment

☒ I consent to examination and treatment

Submit

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REPORT:

Patient Appointment Booking

PAB

Patient Appointment...

Appointment_Booking_Form

Appointment_Booking_Form

Appointment_Booking_Form Report

LIPIKA J 240...

Appointment_Booking_Form Report

Patient_Name	Email Address	Age	Gender	Contact Number	Residential Address	Referred By
LIPIKA J	lipika.j.2024.cse@rajalakshmi.edu.in	20	Female	+919080225657	19/2B Grace garden Royapuram	Dr.Shreyesh
Moorthy S	moorthy.s.2000@gmail.com	60	Male	+919840089133	No 16/7 Richhy street Parrys	Dr.Aasha
PRIYA E	priya.e.2005@gmail.com	45	Female	+919884674602	No.3 Sri Temple street Mylapore	Dr.Rajesh

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WORKFLOW:

creator.zoho.in/appbuilder/lipika.j.2024.cse_rajalak421/patient-appointment-booking/workflow/edit

Patient Appointment Bo...Workflow

DesignWorkflowSettingsUpgradeAccess this application

Form workflowsSchedulesApprovalsPaymentsBlueprintsBatch workflowsFunctions

Search for Workflows or Acti...New Workflow

Search for Forms

Forms with workflows

Appointment_Booking_Form

Workflow Name	Details	Status	Created On
Created > Successful form submission			
Appointment Booked	1 Action	Enabled	23-Sep-2026

5. Result

The survey on emerging trends in low-code application development was **completed successfully**. The low-code development identified major trends such as AI-assisted development, workflow automation, cloud-based development, API integration, rapid prototyping, mobile development. A comparison of existing low-code platforms such as **Zoho Creator**, Microsoft Power Apps, OutSystems, Mendix, and Salesforce Platform was also studied based on their development approach, workflow automation, database support, API integration, AI features, custom coding, and major strengths.

The study concludes that **low-code platforms help reduce development effort and time** while providing visual tools, automation, integration, and reusable components for building modern applications.